DawaDrishti
दवा दृष्टि · India's medicine-quality memory

SOLUTION DOCUMENT · COMMIT FOR GOOD 2026 · AI FOR RESILIENT AND SUSTAINABLE SUPPLY CHAINS

Most failed medicines come from factories we already knew about.

DawaDrishti turns nine years of CDSCO alerts into one national quality memory, with a lookup that any buyer's ERP can call at purchase order and at goods receipt. It returns the factory's public record and a graded answer on what to do now, and every claim is tested against a baseline.

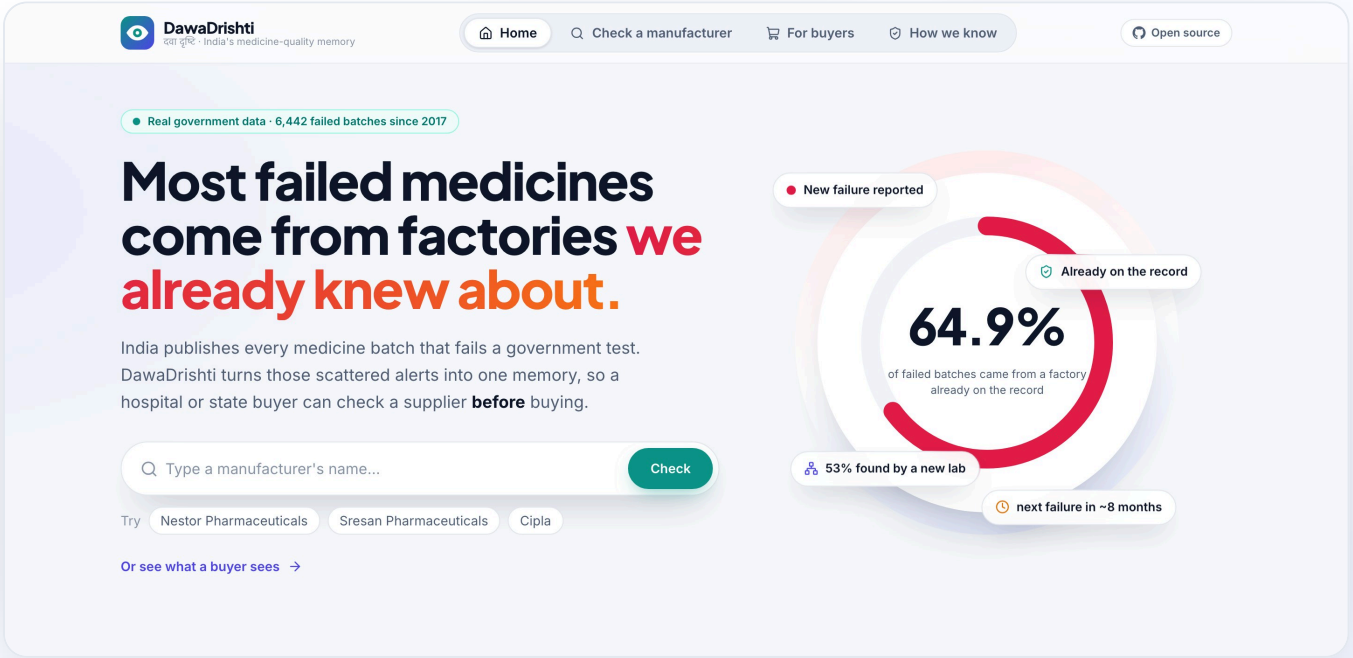


6,442
failed batches studied, Sep 2017 – Aug 2026 (real CDSCO alerts)

8 months
typical gap between a factory's first failure and its next

53%
of repeats found by a lab that had never caught that factory

4.2%
what you'd see with exact names. Name matching is the whole game.



Live: dawadrishti.onrender.com · **Code:** github.com/AbhishekJadli007/dawadrishti (MIT)

Abhishek Jadli · October 2026

IN ONE PAGE

Executive summary

Public medicine buyers in India (state medical-services corporations, hospitals, the Jan Aushadhi scheme) cannot see a supplier's national quality record before they buy, because CDSCO's failure alerts are scattered across monthly PDFs and a portal, the same factory is spelled hundreds of ways, and nothing connects them to buyers' purchasing systems. As a result, repeat failures reach patients: 64.9% of failed batches come from factories already on the national list.

THE PROBLEM

India declared **3,104** drug samples not of standard quality in 2024–25. The worst cases kill: **24 children** had died after taking Coldrif cough syrup by 15 October 2025. CDSCO publishes every failure, but as monthly PDFs and a portal no buyer's system reads. A supplier that fails in one state can keep winning orders in another.

WHAT WE FOUND

We joined nine years of alerts by factory. **64.9%** of failed batches (95% CI 62.2%–67.4%) came from a factory that already had an alert in an earlier month, usually about 8 months earlier, and half the time found by a different lab or state. A buyer checking the record would have seen it.

WHAT WE BUILT

A **national quality memory** and a **lookup API** any ERP can call at purchase order and goods receipt. The public view shows the published record only. The buyer view adds a graded response (pause / test every batch / prefer alternatives). A reference buyer console shows it in use: an order is warned, and a batch already on the national failure list is held on arrival.

HOW WE KNOW IT WORKS

Data: the PDF reader agrees with CDSCO's portal on 96.9% of rows.

Names: name matching made no false merges on held-out pairs.

Prediction: the model beats the best simple rule on AUC (0.774 vs 0.745), and we report where the rule wins.

Process checks: the same engine reproduces the published counts of a 1.6M-event SAP log.

Try it in two minutes

1. Open dawadrishti.onrender.com. Let it wake up (about 1 minute, free hosting).
2. **Check a manufacturer** → "Nestor Pharmaceuticals" → switch to **Buyer view**.
3. **For buyers** → "Use the demo order" → order from Scott-Edil → use the red batch chip → watch it get held.

Which AI capabilities, and why

Capability (from the brief)	In DawaDrishti	Why this method
Explain	Entity resolution and the national record per factory	Fuzzy matching with a human review queue: transparent, testable, no black box over named firms
Detect	Exact-batch check at goods receipt; conformance rules on event logs	Deterministic rules where the answer must be certain
Recommend	Graded response with concentration and days of cover	Simple rules a drug controller can audit, with the supply cost shown beside them
Predict	Future regulatory attention; batch test priority	Gradient boosting with a time split, always next to baselines
Ask	Plain-English questions → one of nine reviewed query templates	Gemini only picks a template; it never writes SQL or text about a firm

STEP 1–2 · OBSERVE AND INVESTIGATE

Where the medicine supply chain breaks

The participant guide splits a supply chain into source, make, store, move, distribute and use, with information travelling alongside the goods. For public medicines, the goods flow works: factories make, buyers order, warehouses store, hospitals dispense. The **information flow** is what fails. When a government lab finds a bad batch, the warning is published, but it does not travel to the next buyer's purchase order.



Who is affected, and how

Patients

Receive medicine that is weak, does not dissolve, is contaminated or fake. In the worst cases, like diethylene glycol in cough syrup, children die.

Public buyers

State medical-services corporations, district hospitals and Jan Aushadhi procure through tenders. They test their own stock, but cannot see failures found in other states.

Regulators

CDSCO and state drug controllers publish failures and, since October 2025, run risk-based inspections. They need to know where to sample next.

Honest manufacturers

Compete on price against firms with repeat failures, and must not be confused with a similarly named firm.

How a failure travels today

1. A state lab in Kerala tests a sample of tablets and finds they do not dissolve. The result goes to CDSCO.
2. Weeks later it appears in CDSCO's monthly list: one row among about 200, with the maker's name typed by hand.
3. A corporation in another state opens a tender for the same medicine. Its ERP (DVDMS, e-Aushadhi or its own) checks price and stock, **not** the national record. The same factory wins again. Its next batch fails about 8 months later, found by yet another lab.

What each group needs, and what DawaDrishti gives them

Who	Needs	DawaDrishti gives
Buyer	Know a supplier's record before ordering, in seconds, inside their own system	One API call at purchase order and goods receipt; a clear "what to do now"
Regulator	Decide where to sample next with limited inspectors	Units ranked by future regulatory attention, tested against simple rules
Manufacturer	Accuracy, and not being confused with a similarly named firm	Evaluated name matching, no guessed identities, a correction route
Patient	Medicine that works	Fewer repeat failures reaching the shelf; a batch on the national list held on arrival

STEP 3–4 · QUANTIFY AND CHALLENGE

The evidence, and why existing systems miss it

Fact	Detail	Source
3,104 samples NSQ	declared not of standard quality in 2024–25, out of 1.16 lakh tested (about 2.7%)	Lok Sabha via IndiaMedToday
211 in one month	samples flagged NSQ by CDSCO labs in October 2025 alone	IANS
24 children	had died after taking Coldrif syrup (diethylene glycol), as of 15 October 2025	ETV Bharat
66 children	deaths in The Gambia linked by WHO to syrups from an Indian manufacturer (2022)	The South First (WHO alert)
40 failed samples	from one supplier in Rajasthan's state testing (2021–23). It was banned for a year, then reinstated	Down To Earth; Deccan Herald
6,442 failed batches	in CDSCO's own alerts, Sep 2017 – Aug 2026, which we extracted, cleaned and joined (our data)	CDSCO portal and PDFs

What already exists, and the gap

Existing	What it does	Where it falls short
CDSCO NSQ alerts	Publishes every failed batch monthly (PDFs since 2017, portal since 2019)	A list, not a memory: no factory identity, no history per firm, no API, one factory spelled up to 14 ways
DVDMS / e-Aushadhi	Buyers' ERPs: purchase, stock, distribution, quality control	Never read the national alerts. Each state's failures stay in its own system
Risk-based inspection	CDSCO began targeting factories in 6 states after the 2025 deaths	Regulator-side only. Says nothing to the buyer about to place an order
Buyers' own testing	Quarantine and test each batch before issue	Slow labs create pressure to release untested stock, and a buyer's own test can't see failures found elsewhere

The gap in one line: the evidence exists and is public, but it is not joined up by factory, not trusted enough to act on, and not available at the moment of purchase.

Why this matters for resilience

The guide names the recurring failure modes. This problem is three of them at once: **information arrives late, incomplete and in incompatible formats**; **local decisions ignore the rest of the network**, because a buyer can't see another state's failure; and **one supplier can be a single point of failure**, so pausing a bad supplier must be weighed against supply. DawaDrishti addresses the first two directly, and shows the third (concentration and days of cover) at the moment of decision.

Why now

The data became machine-readable

CDSCO's portal has carried structured alerts since 2019, alongside the PDFs. For the first time the whole record can be joined.

The buyers went digital

States now run DVDMS / e-Aushadhi. There is finally a system to plug the lookup into at the moment of purchase.

The regulator moved

After the 2025 deaths, CDSCO started risk-based inspection. The same signal can now reach buyers, not only inspectors.

STEP 5 · DEFINE, AND TEST IT WITH DATA

The problem in one sentence, and our finding

WHO
Public medicine buyers: state medical-services corporations, hospitals, Jan Aushadhi

CANNOT
see a supplier's national quality record before they buy

BECAUSE
alerts are scattered PDFs, one factory has many spellings, and nothing reaches their ERP

WHICH CAUSES
repeat failures to reach patients: 64.9% of failed batches come from factories already listed



What we measured

For every failed batch we asked: had the same factory appeared in an alert in an **earlier month**? Measured on 6,442 batches from 1,914 factories. The 95% confidence interval (62.2%–67.4%) comes from resampling factories 1,000 times. Among critical failures (contamination, sterility, adulteration, fakes) the share is 59.0%.

8 mo
median gap from first failure to the next

53%
of repeats found by a lab new to that factory

53%
of repeats found in a state new to that factory

48%
of all failures from 10% of factories (Gini 0.55)

Does it depend on how names are matched?

Exact strings	4.2%
Normalised names	62.8%
DawaDrishti (default)	64.9%
Factories joined across states	67.5%

Above exact strings, every rule gives 63–68%. Missed merges lower the share, so ours is conservative.

The honest counter-finding

Case	Alerts before	Total
Sresan Pharmaceutical (Coldrif)	0	11
Maiden Pharmaceuticals (The Gambia)	0	6
Kaysons Pharma (Rajasthan)	0	0

The national alerts alone would not have warned anyone about these three. So the memory must also take **state-lab and buyer results**, which is the next step on our roadmap.

Step 6 · Build: can AI realistically help?

Yes, but mostly with careful data work, not a big model. The hard part is **knowing which names are the same factory**, at precision high enough to name firms safely. Then come simple, auditable rules at the moment of decision. Prediction helps regulators decide where to sample, and it is held to the standard of beating simple rules.

THE SOLUTION

A national quality memory, and a lookup every buyer can call

DawaDrishti is **not another ERP** and **not a blacklist**. It is the layer that has to exist before a shared ERP would be worth anything. It is a public-record aggregator with a risk signal for buyers, and it plugs into the systems buyers already run.

Public

Anyone

Look up a manufacturer. See every published failure, with month, medicine, test failed, lab and batch. Also shown: the as-of date, what the data can't tell you, and how to request a correction. No predictions.

Buyer

Procurement officer

At purchase order: a graded response with supply concentration beside it. At goods receipt: a batch already on the national list is held at once. Issue stays blocked until the lab clears the batch.

Regulator

Drug controller

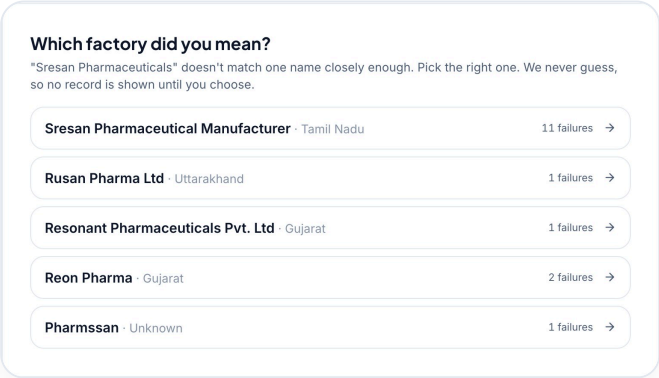
Where to sample next: units ranked by future regulatory attention, with the alerts behind each rank. A conformance engine checks buyers' event logs for stock released before testing.

The graded response (same rules in the API and in the console)

Level	Response	When	In the buyer's ERP
3	**Pause new orders**	A critical failure in the last 12 months, or debarred by a buyer	PO blocked unless an override is recorded
2	**Test every batch**	A critical failure 13–36 months ago, or 2+ alerts in 12 months	No skip-lot testing
1	**Prefer alternatives**	Any alert in the last 36 months	At equal price, choose another supplier
0	**No recent record**	Nothing in 36 months	Absence of a record, not a certificate

The link is the factory, and the batch is a bonus

Alerts carry the manufacturer's batch numbers and no buyer. So a buyer's own stock matches an alert by batch only when it bought that exact batch. That is rare, but decisive: the batch is held on arrival. Everything else works at factory level, which is where the 64.9% signal lives.



A buyer's journey with DawaDrishti

1

Raise a purchase order
The ERP calls the lookup for each supplier on the rate contract.

2

See the graded response
Traffic lights per supplier, with how concentrated supply is.

3

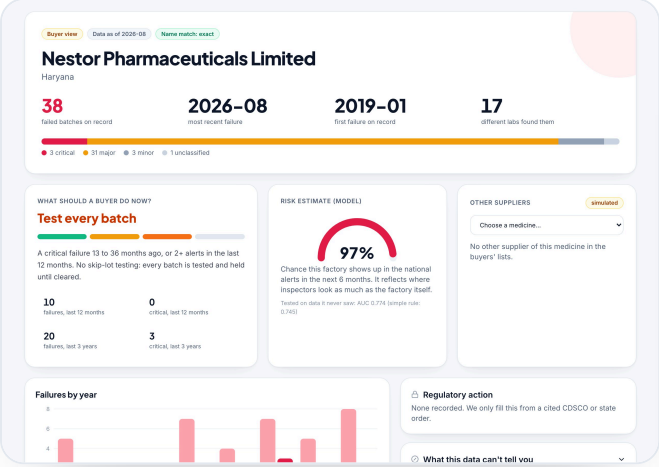
Receive the goods
The batch number is checked against the national list; a match is held.

4

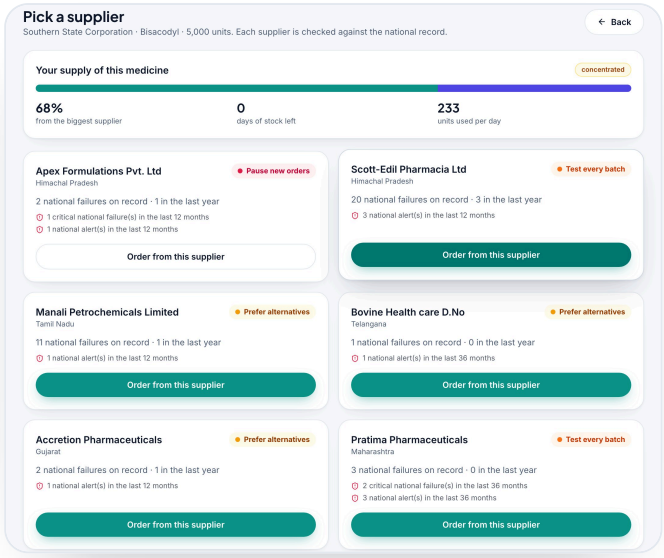
Release only when cleared
Issue is blocked until the lab clears the batch. Every step is logged.

PRODUCT TOUR · LIVE AT DAWADRISHTI.ONRENDER.COM

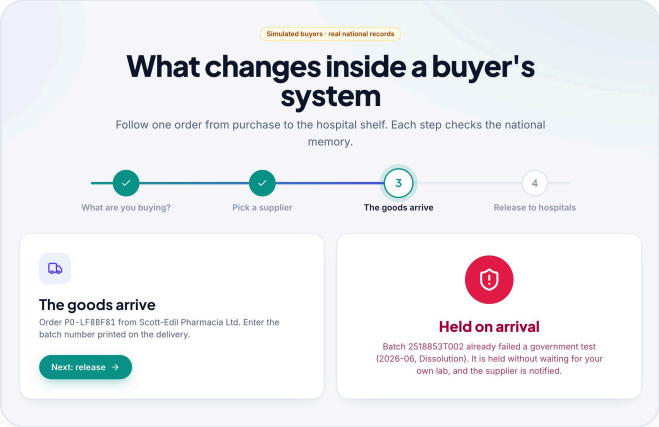
What it looks like



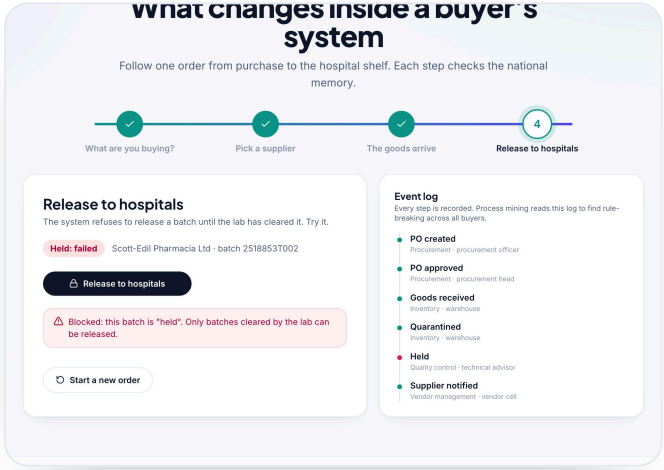
Check a manufacturer, buyer view. 38 published failures; the graded response is "Test every batch"; the model's estimate is labelled with its tested accuracy.



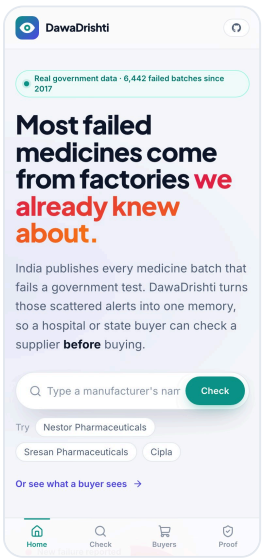
For buyers, step 2. Suppliers on the rate contract, each with its traffic-light response. Supply concentration and days of cover shown first.

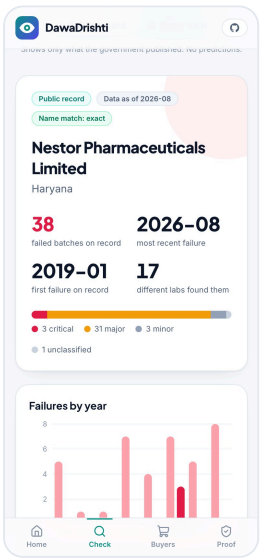


Goods receipt. This exact batch already failed a government test, so it is held on arrival and the supplier is notified.



Release. Issue to hospitals is refused until the lab clears the batch. Every step lands in the event log that process mining reads.

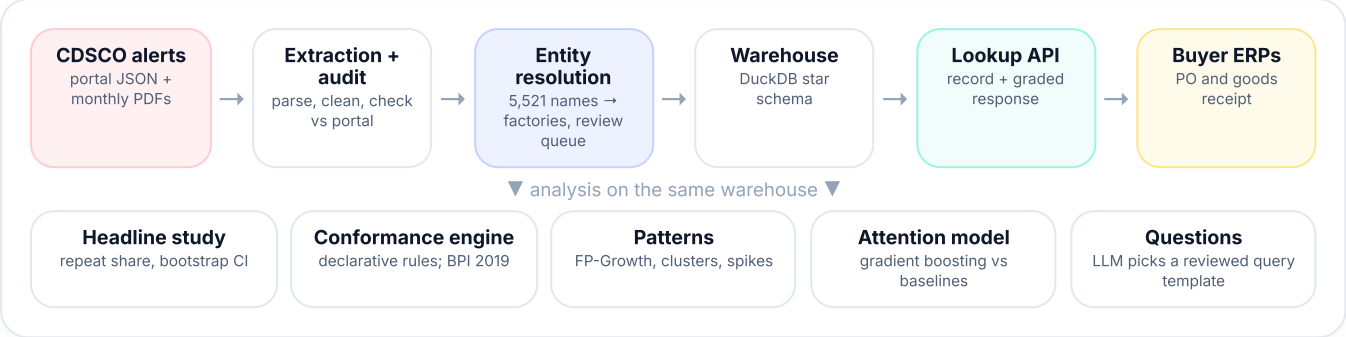




Designed for non-experts: four sections (Home, Check, For buyers, How we know), plain language, motion that explains (the ring fills, numbers count up, a blocked release shakes), and it works on a phone. The **How we know** section holds every piece of evidence: the study, data quality, models, process checks, patterns, and a question box.

HOW IT WORKS

Architecture



Component	Method	Checked by
Extraction	Portal JSON (2019–26) and pdfplumber for PDFs (2017–18). Rule-based cleaning of name, address, state (last place named wins), product, failure type and severity	Audit against the portal: 96.9% of rows match; fields agree 96.7–99.8%
Entity resolution	Normalisation rules, blocking, fuzzy scoring, union-find. Unit = firm + state; firm = units joined across states. Score 80–92 goes to a person	200 labelled pairs (120 development, 80 held out); threshold sweep
Warehouse	DuckDB star schema. Two fact tables (failures, buyer events) share supplier, product and date dimensions	Rebuilt from source by one command
Lookup API	FastAPI. Name resolution (candidates below the threshold), record, graded response, exact-batch check, alternatives	Policy tests: no score in the public view; key-gated buyer view
Conformance engine	Declarative rules (precedence, response within N days, requires) plus process discovery. Written from scratch, because pm4py is AGPL	Identical to hand-written checks; BPI 2019 reproduced
Models	Gradient boosting with a time split: future regulatory attention (real), batch test priority (simulated)	AUC and top-k against simple rules; confound checks
Questions	Gemini (or keyword rules) picks one of nine SQL templates; parameters are validated against the data and bound	Injection tests; never writes SQL
Interface	React, Tailwind, framer-motion, Recharts and Leaflet, served by the API in one Docker image	Phone and desktop checks; 22 automated tests

5,521 → 1,994
name strings resolved into factories (1,797 firms)

~190 MB
memory in production: runs on a free Render instance

1 command
`python -m dawadrishti build`
rebuilds everything from the public data

EVALUATION

Every claim, against a baseline

Question	Data	Result	Against what
Failed batches from factories already on record	Real	**64.9%**	95% CI 62.2%–67.4%; exact names would show 4.2%
PDF reader vs CDSCO portal	Real	**96.9%**	rows matched; fields agree 96.7%–99.8%
Name matching, held-out hard pairs	Real	**P 1.00 / R 0.62**	exact names R 0.00; earlier rules R 0.48; with review R 0.98
Who appears in the next 6 months of alerts	Real	**AUC 0.774**	best simple rule 0.745; top-50: simple rule wins (84% vs 76%)
Same, when a new lab finds the failure	Real	**AUC 0.765**	330 of 386 units; state-only model 0.601
Conformance engine on a real SAP log	Real	**655**	payments before goods receipt in 1,595,923 events; published counts reproduced exactly
Test the riskiest batch first	Simulated	**69.1%**	of failures caught in first 30% of tests vs 27.5% first-in-first-out
Shared memory vs isolated ERPs	Simulated	**−81%**	orders to debarred suppliers; failed batches received −30%

Who will appear in the next 6 months of alerts? (real)

Trained up to Sep 2025, tested on Mar–Aug 2026: 1,737 units, 386 appeared



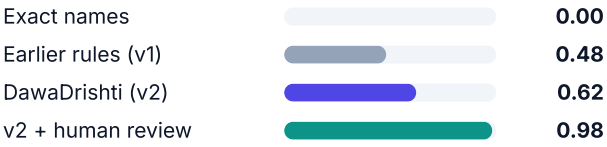
AUC (bars start at 0.4). On the top 50 alone, "most recent failure" wins: 84% vs 76%. We say so.

Is it only learning where inspectors look?

Units scored	Model	Simple rule
units in HP, Uttarakhand, Gujarat (most sampled)	0.758	0.705
units in all other states	0.756	0.763
next failure found by a new lab (330)	0.765	0.721

A model that knows only the state scores 0.601.

Name matching on held-out hard pairs



Recall shown; precision is 1.00 for every automatic rule, so no false merges. Labels were made by the assistant and await human review.

Conformance engine on a log we did not write

BPI Challenge 2019 (SAP)	Published	Ours
events	1,595,923	1,595,923
cases	251,734	251,734
purchase documents	76,349	76,349
activities	42	42

655 invoices paid before goods receipt; **0** where SAP enforces the rule. Enforced rules hold, policies leak, which is why DawaDrishti enforces "no issue before clearance".

How we tested

Time splits for every model: trained only on the past, scored on months it never saw. **Held-out sets** for name matching: test pairs drawn after the rules were fixed, labelled with scores hidden. **Baselines** everywhere: simple rules a regulator or buyer would otherwise use. **Uncertainty** by cluster bootstrap. **Robustness** by recomputing the headline under five matching rules. **Simulated results are labelled** and never presented as measured outcomes.

RESPONSIBLE BY DESIGN

Naming real firms, carefully

DawaDrishti names real manufacturers, so how it names them is part of the design. It is a **public-record aggregator with a risk signal for buyers, not a blacklist**. A blacklist asserts a conclusion. We present evidence that already exists, assembled in one place, with its provenance attached.

Public pages show records, never predictions

The API enforces it: public responses carry no score and no ranked list. The buyer view can require a key.

Every figure is dated

Each answer carries the as-of date and a list of what the data does not contain: passing batches, licence actions, harm, sampling effort.

A correction route on every record

Anyone can ask for a correction or an annotation. A regulatory_action field is filled only from a cited order.

No guessed identities

An unclear name returns candidates to choose from, never one firm's record under another firm's name.

Fakes are not counted against the victim

A fake whose named firm denied making it is excluded (45 rows), and so is a row naming a buyer instead of a maker (35).

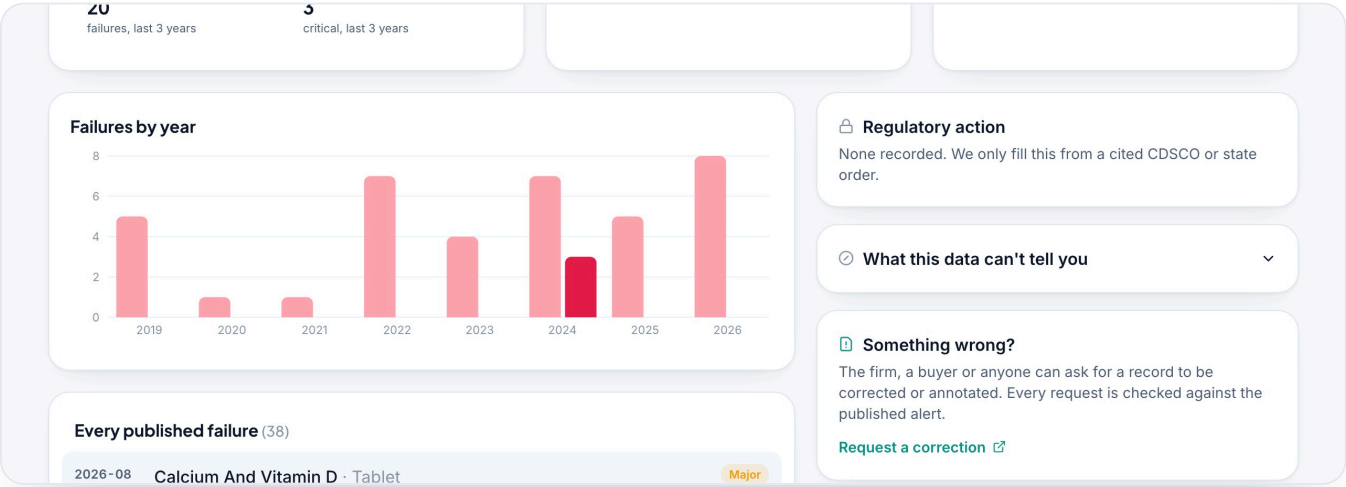
The language model never characterises a firm

It only picks one of nine reviewed query templates. Its parameters are checked against the data before anything runs.

Data, and what is real

Data	Status	Licence / note
CDSCO NSQ alerts 2017–2026 (6,442 used)	Real	Government of India public information; re-downloadable by script
BPI Challenge 2019 SAP event log	Real	4TU.ResearchData; not committed (728 MB), but the result is
Five buyers' ERP history (orders, batches, tests, debarments)	Simulated	Seeded generator, assumptions in docs/ASSUMPTIONS.md; supplier names and records are real
Entity-resolution labels (200 pairs)	Assistant-labelled	Awaiting human review; a blind review tool is included

How it looks on a record



OPEN SOURCE

Try it, run it, extend it

Real input and output

One GET request from any ERP at purchase order and goods receipt. Below is a real response, trimmed.

```
$ curl "dawadrishti.onrender.com/api/v1/lookup?q=Nestor%20Pharmaceuticals%20Ltd&view=buyer&batch=CITB-02&molecule=Ciprofloxacin"
{
  "identity": {"manufacturer_id": "M00008", "match": "exact", "score": 100.0},
  "record": {"unit": {"manufacturer_id": "M00008", "name": "Nestor Pharmaceuticals Limited", "state": "Haryana"},
"alerts": 38, "first_alert": "2019-01", "last_alert": "2026-08", "by_severity": {"major": 31, "minor": 3, "critical": 3,
"unclassified": 1}, "entries": [{"month": "2026-08", "generic_name": "Calcium And Vitamin D", "dosage_form": "Tablet",
"primary_failure": "Dissolution", "severity": "major", "reporting_lab": "RDTL, EKM", "reporting_source": "State lab",
"batch_no": "CUTC-39", "is_spurious": false, "state": "Haryana"}], "regulatory_action": null},
  "batch_check": {"batch": "CITB-02", "this_batch_failed": true},
  "buyer": {"graded_response": {"level": 2, "label": "Test every batch"}, "future_regulatory_attention":
{"probability_next_6_months": 0.965}},
  "data_as_of": "2026-08",
  "data_does_not_contain": ["Batches that were tested and passed. The alerts list failures only, so a failure rate
cannot be computed.", "Licence suspensions, cancellations, prosecutions or court outcomes. The regulatory-action field
is filled only from a cited order.", "…"]
}
```

Open-source checklist

Licence	✓ MIT
README: run steps, example, limitations	✓ yes
Architecture, evaluation, assumptions	✓ docs/
Data sources and licences	✓ ACKNOWLEDGEMENTS.md
CONTRIBUTING + correction template	✓ yes
Automated tests	✓ 22 passing
One-command rebuild from public data	✓ yes
Docker image, live URL	✓ Render (free)
Secrets kept out of the repo	✓ .env.example
v1.0 tag and release	✓ GitHub

Run it yourself

```
git clone
https://github.com/AbhishekJadli007/dawadrishti.git
cd dawadrishti
docker build -t dawadrishti .
docker run -p 8000:8000 dawadrishti
# open http://localhost:8000
```

Where help is most useful

- Reviewing the 200 name-matching labels (a blind review tool is included)
- A route for state labs and buyers to send results in
- A DVDMS / e-Aushadhi connector that calls the lookup
- Product mapping to WHO ATC and India's NLEM

LIMITS, NEXT STEPS, SOURCES

What this does not do yet

- **Alerts list failures, not tests.** No failure rate per factory, so we measure repetition on the record.
- **Sampling is uneven.** The prediction is of regulatory attention, not true quality, and is never shown publicly.
- **The national list misses state findings.** Coldrif, Maiden and Kaysons had no prior national alert. That needs the contribution route below.
- **Name-matching labels** are the assistant's until people review them. Recall on the hardest pairs is 0.63; missed merges make the headline conservative.
- **Buyer-side results are simulated.** The network effect and batch priority show the method, not a measured outcome.

Roadmap

Next 3 months

Human review of the labels. A contribution route for state labs and buyers to send failures in. A correction workflow.

6 months

Pilot with one state medical-services corporation: call the lookup from its ERP at PO and goods receipt; measure orders changed.

12 months

DVDMS / e-Aushadhi connector. A licence master for true alternatives. Real buyer event logs into the conformance engine.

Team

Abhishek Jadli (@AbhishekJadli007). Solo entry: data pipeline, entity resolution, study, API, conformance engine, models, interface, deployment and documentation. An AI assistant (Claude) helped with research and code, every part reviewed. See ACKNOWLEDGEMENTS.md.

Sources

- CDSCO NSQ alerts: public portal and monthly PDFs
- Lok Sabha via IndiaMedToday: 3,104 of 1.16 lakh samples NSQ in 2024–25
- IANS: 211 samples flagged NSQ in October 2025
- ETV Bharat, 15 Oct 2025: Coldrif death toll reaches 24
- The South First: WHO alert, syrups linked to 66 child deaths in The Gambia
- Down To Earth: 40 failed samples from one supplier in Rajasthan
- Deccan Herald: Rajasthan halts Kaysons distribution, suspends drug controller
- DT Next: CDSCO starts risk-based inspection in 6 states
- BPI Challenge 2019 (van Dongen, 4TU.ResearchData)

Find the problem. Prove it matters. Build the solution. Commit it to open source.

Live: dawadrishti.onrender.com · Code: github.com/AbhishekJadli007/dawadrishti · Licence: MIT